**FINDING TIME COMPLEXITY OF ALGORITHMS**

Problem 1: Finding Complexity using Counter Method

```
#include<stdio.h>
int main(){
    int n;
    scanf("%d",&n);
    int i=1;
    int s=1;
    int count=0;
    count++;
    count++;
    while(s<=n){
      count++;
      i++;
      count++;
      s+=i;
      count++;
  }
  count++;
  printf("%d",count);
 return 0;
}
```

Problem 2: Finding Complexity using Counter method

```
#include <stdio.h>
int main() {
    int n, count = 0;
```

```
    scanf("%d", &n);
    if (n == 1) {
        count += 2;
    } else {
        count++;
        for (int i = 1; i <= n; i++) {
            count += 5;
        }
        count++;
    }
    printf("%d", count);
    return 0;
}
```

Problem 3: Finding Complexity using Counter Method

```
#include<stdio.h>
int main(){
    int num;
    scanf("%d",&num);
    int count=0;
    for(int i=1;i<=num;++i){
        count++;
        count++;
        if(num%i==0){
            count++;
        }
    }
```

```
    count++;
    printf("%d",count);
    return 0;
}
```

Problem 4: Finding Complexity using Counter Method

```
#include <stdio.h>
int main() {
    int n;
    scanf("%d", &n);
    int count=0;
    count++;
    for (int i = n / 2; i < n; i++){
        count++;
        for (int j = 1; j < n; j =2*j){
            count++;
            for (int k = 1; k < n; k = k*2){
                count++;
                count++;
            }
                count++;
        }
        count++;
    }
    count++;
    printf("%d", count);
    return 0;
}
```

Problem 5: Finding Complexity using counter method

```
#include<stdio.h>
int main(){
    int n;
    scanf("%d",&n);
    int rev=0;
    int remainder=0;
    int count=0;
    count++;
    count++;
    while(n!=0){
        count++;
        remainder=n%10;
        count++;
        rev=rev*10+remainder;
        count++;
        n/=10;
        count++;
    }
    count++;
    printf("%d",count);
    return 0;
}
```